

Construct Validity for Geometric Perturbation Scores: When Does a Number License a Mechanistic Claim?

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Abstract

Background: Geometric perturbation scores—such as direction instability, which measures the angular consistency of perturbation signatures across biological contexts—are used to rank drugs and genes by mechanism conservation. These scores predict biological properties, but what a given score *licenses* as a mechanistic claim depends on validity conditions that the raw number does not encode. We propose a construct-validity framework (a “validity ladder”) that specifies which interpretive threats a score has survived, and test whether these conditions matter empirically.

Methods: We conducted pre-registered experiments across three perturbation domains: 8,949 LINCS L1000 drug compounds, 1,676 Perturb-seq CRISPRi gene knockdowns, and 25,254 JUMP-CP Cell Painting morphological profiles. Five drug hypotheses and three genetic hypotheses were tested with Holm–Bonferroni correction within each domain family. Validity conditions included phenotype projection onto genetic perturbation directions, transport-stable variance correction with leave-one-out cross-validation, and essential-gene confound correction on Grassmannian subspaces.

Results: Domain-specific validity conditions preserved global rankings in all three domains (LINCS: $\rho > 0.99$; Perturb-seq: $\rho = 0.91$; JUMP-CP: $\rho = 0.99$) while upgrading inferential warrant at the item level. Phenotype-projected scores correlated with on-target genetic perturbation connectivity ($\rho = 0.38$, $p < 10^{-27}$) while raw scores did not ($\rho = -0.04$). Transport-stable scores outpredicted raw scores for held-out cell-line effects in all 66 valid leave-one-out folds (Wilcoxon $p = 8.2 \times 10^{-13}$; mean $\Delta\rho = +1.10$, 95% CI [1.00, 1.19]). Essential-gene correction revealed that universal-essential knockdowns *diverge* rather than converge across cell types ($d = 0.60$, $p_{\text{perm}} < 10^{-4}$), inverting the predicted confound. Of 5 drug hypotheses, 2 survived multiplicity correction, 1 was refuted, and 2 were informative nulls.

Conclusions: Validity conditions do not change geometric perturbation scores; they change what a practitioner can conclude from them. The construct-validity framework tested here—applicable to any geometric score that measures perturbation-response consistency across contexts—provides a structured procedure for accumulating interpretive evidence, separating scores that license mechanistic claims from scores that merely rank.

1 Introduction

Geometric perturbation scores—metrics that quantify how a biological system’s response to a perturbation varies across experimental contexts—are used throughout computational biology to rank drugs, genes, and compounds by mechanism conservation [5, 12]. Direction instability, one such score, measures the angular consistency of perturbation signatures across cell lines: a drug whose transcriptomic response points in the same direction regardless of cell type receives a low score, indicating conserved mechanism [14]. Applied to 8,949 drugs in LINCS L1000, direction instability produces biologically meaningful rankings: pan-HDAC inhibitors achieve the lowest scores ($D = 0.14$ – 0.19), with a selectivity gradient within the HDAC class (Figure 3) that tracks isoform specificity [13].

These rankings establish that the score captures genuine biological signal. They leave open a different question: under what conditions does a geometric perturbation score license a mechanistic *claim*? A score

of $D = 0.55$ means signatures point in similar directions across cell lines. It does not mean the drug acts through a specific pathway or that the observation will replicate in a new context.

This gap between score and claim is a construct-validity problem [7]. In Messick’s framework, a score’s meaning is not inherent in the number; it depends on accumulating evidence from multiple sources that the score measures what it purports to measure and that interpretations drawn from it are warranted. The same principle applies to geometric perturbation scores: a raw ranking may be reproducible and biologically correlated, yet insufficient to license mechanistic inference without additional validation evidence.

Three empirical threats illustrate this gap. First, a drug’s low direction instability could reflect a coherent off-target program rather than on-target mechanism conservation—the raw score cannot distinguish the two. Second, a drug with low direction instability in training data may fail to predict a held-out cell line’s response, because the raw metric conflates genuine directional consistency with Fréchet variance from a small number of contexts. Third, in genetic perturbation screens, universal-essential knockdowns could inflate or deflate cross-cell-type distances through a shared lethal-response subspace, contaminating transport scores that appear mechanistically informative.

We formalize these threats through a validity ladder (Figure 2): a sequence of seven conditions that progressively strengthen what a geometric perturbation score licenses. Each rung answers a specific threat to interpretation. The ladder is a domain-specific instantiation of construct validation, analogous to the Bradford Hill criteria for causal inference [4]: satisfying more rungs does not change the number, it changes what a practitioner is warranted in concluding from it. The rung ordering reflects logical dependency—a reliable measurement (Rung 2) is prerequisite for meaningful projection (Rung 5), and localized signal (Rung 4) precedes phenotype alignment (Rung 5)—rather than a strict hierarchy of evidence strength.

This paper tests whether validity conditions matter empirically across three perturbation domains: 8,949 LINCS L1000 drug compounds [12], 1,676 Perturb-seq genetic knockdowns [10], and 25,254 JUMP-CP Cell Painting morphological profiles [2]. Direction instability serves as the case study, but the construct-validity procedure—define threats, specify corrections, pre-register predictions, test whether corrections reorganize rankings or preserve them—applies to any geometric score measuring perturbation-response consistency across contexts. We report honest nulls alongside confirmations, because the pattern of where validity conditions matter—and where they do not—is itself the finding.

2 Background

2.1 Direction instability

For a drug tested in K cell lines, let $\mathbf{s}_1, \dots, \mathbf{s}_K \in \mathbb{R}^{978}$ denote its consensus signature vectors. Define unit-normalized signatures $\hat{\mathbf{s}}_i = \mathbf{s}_i / \|\mathbf{s}_i\|_2$. Direction instability is:

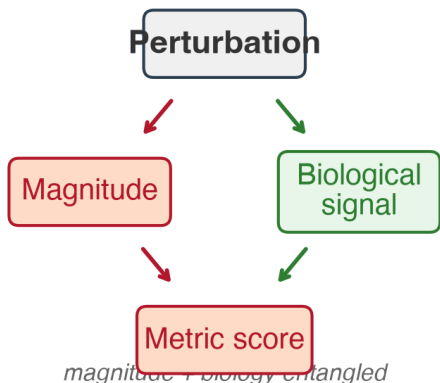
$$D = 1 - \frac{2}{K(K-1)} \sum_{i < j} \hat{\mathbf{s}}_i^\top \hat{\mathbf{s}}_j \quad (1)$$

$D = 0$ when all signatures are collinear. $D = 1$ when orthogonal on average. Cosine normalization renders direction instability invariant to perturbation magnitude (Figure 1): variance-based and estimation-based metrics conflate perturbation strength with biological signal, but direction instability measures only direction [14].

2.2 Grassmannian geodesic distance in Perturb-seq

Direction instability generalizes from vectors to subspaces. For a gene knocked down in two cell types, let $U_1, U_2 \in \text{Gr}(k, d)$ denote the top- k PCA subspaces of the perturbation response in each cell type (k = subspace dimension, d = ambient dimension). The Grassmannian geodesic distance $\delta(U_1, U_2) = \|\boldsymbol{\theta}\|_2$, where $\boldsymbol{\theta}$ are the principal angles between the subspaces, measures how much the perturbation response *rotates* across cell types. With $K = 2$ contexts, this reduces to a single pairwise comparison—a weaker geometric estimand than the $K \geq 5$ angular-consistency computation used in LINCS and JUMP-CP, as it carries no Fréchet variance or holonomy information. We interpret the Perturb-seq results accordingly.

a Confounded metrics



b Direction instability

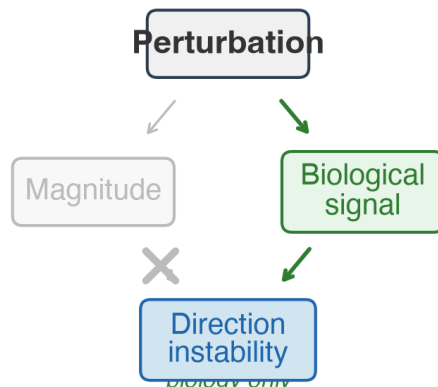


Figure 1: **Direction instability is invariant to perturbation magnitude.** (a) Confounded metrics (variance, subspace size) inherit effect magnitude through a shared path, producing spurious associations between perturbation strength and metric value. (b) Direction instability operates after cosine normalization, which renders the metric invariant to perturbation magnitude. The observed association between group variable and metric reflects biological signal only. This invariance is necessary but not sufficient: direction instability still requires validity conditions to distinguish on-target from off-target consistency (Rung 5) and genuine transport from sampling noise (Rung 6).

2.3 Direction instability in Cell Painting morphological profiles

Cell Painting [2] measures 3,180 morphological features per well across five fluorescent channels. JUMP-CP profiles 116,000 compounds across 10 imaging sources. Direction instability applies directly: each compound’s morphological signature in a given source is a vector in $\mathbb{R}^{3180}$, and D measures angular consistency across sources. A subset of 743 features—cell area, compactness, eccentricity, DNA/Mito intensity, granularity, and radial distribution—reflect general cell health rather than compound-specific morphological effects. The cell-health correction removes these features before recomputing direction instability.

2.4 The validity problem

Consider two drugs with identical direction instability, $D = 0.55$:

- **Drug A** is a pan-HDAC inhibitor. Its conserved direction aligns with its target gene’s knockdown signature—phenotype-projected bracket is high. The consistency comes from on-target chromatin remodeling.
- **Drug B** is an off-target consistent compound. Its conserved direction does *not* align with its annotated target’s knockdown signature—phenotype-projected bracket is near zero. The consistency comes from an off-target program shared across cell types.

Both achieve $D = 0.55$. Both pass the raw metric equally. Phenotype projection (Rung 5 of the validity ladder) distinguishes them: Drug A’s consistency is mechanistically on-target; Drug B’s is not. The distinction matters for downstream decisions: Drug A is a strong candidate for mechanism-based repurposing; Drug B requires additional investigation to identify what program drives its consistency before any mechanistic claim is warranted.

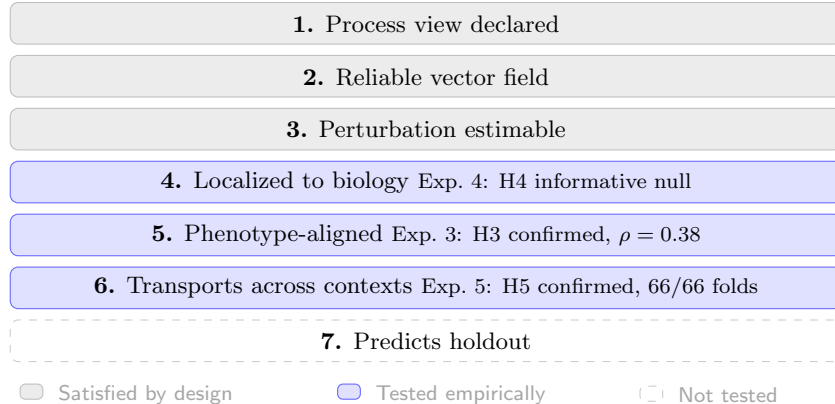


Figure 2: **The validity ladder.** Seven rungs progress from measurement prerequisites (1–3) through empirically tested validity conditions (4–6) to external prediction (7). Each rung eliminates a specific interpretive threat. Rungs 4–6 are tested across three domains (LINCS L1000, Perturb-seq, JUMP-CP); result summaries are shown at right. Satisfying additional rungs does not change the direction-instability score—it upgrades the inferential warrant attached to it.

2.5 The validity ladder

The validity ladder is a domain-specific instantiation of Messick’s construct-validation framework [7], adapted for geometric perturbation scores. We define seven rungs (Figure 2), ordered by logical dependency rather than by evidence strength:

1. **Process view declared.** What constitutes a “context”?
2. **Reliable vector field.** Adequate signal-to-noise for direction to be meaningful.
3. **Perturbation estimable.** Effect isolated from background.
4. **Localized to biology.** Instability concentrates in pathway-relevant genes.
5. **Phenotype-aligned.** Conservation direction matches a known target pathway.
6. **Transports across contexts.** Score replicates in held-out contexts (low Fréchet variance).
7. **Predicts holdout.** Metric predicts an external outcome in new data.

Rungs 1–3 are satisfied by design in our datasets and are not tested further. Rungs 4–6 are the validity conditions we test empirically. Rung 7 (holdout prediction in a genuinely new dataset) is not tested: Experiment 5’s leave-one-cell-line-out cross-validation tests Rung 6 (transport within LINCS), not generalization to an independent dataset.

3 Related work

Construct validity in measurement. Messick’s unified validity framework [7] established that a score’s meaning depends on accumulating evidence across multiple sources—content, substantive, structural, generalizability, external, and consequential. Our validity ladder instantiates this framework for geometric perturbation scores: Rungs 1–3 address content and structural validity, Rungs 4–5 address substantive and external validity, and Rung 6 addresses generalizability. Recent construct-validity frameworks for computational scores [4, 15] and the SBB principles for perturbation-prediction models [11] share this orientation.

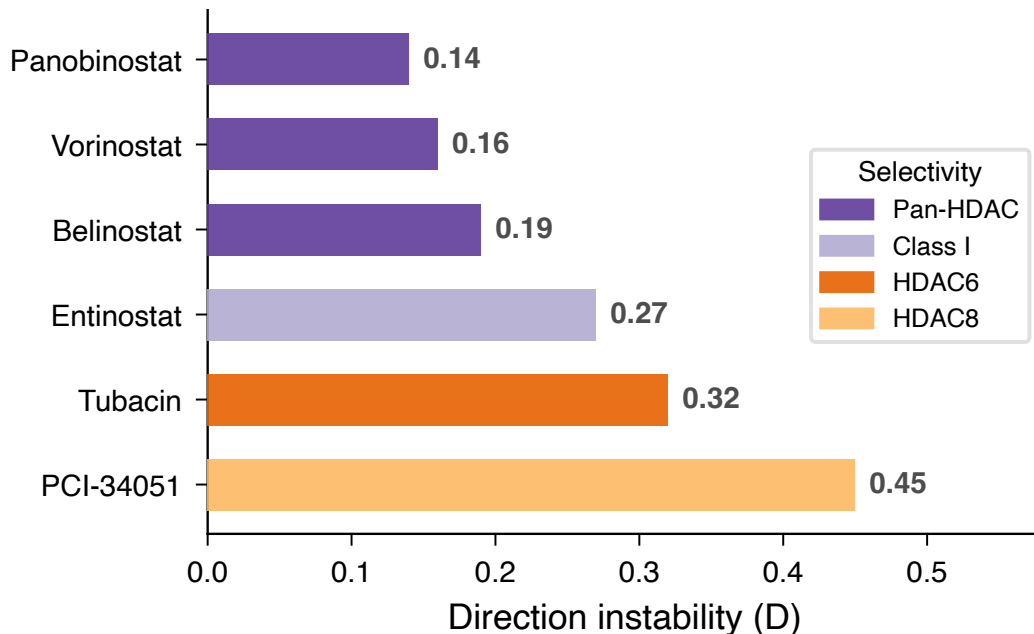


Figure 3: **HDAC selectivity gradient: calibration standard for direction instability.** Pan-HDAC inhibitors (panobinostat, vorinostat, belinostat) achieve the lowest direction instability ($D = 0.14$ – 0.19); Class I-selective entinostat is intermediate ($D = 0.27$); isoform-selective HDAC6 and HDAC8 inhibitors (tubacin, PCI-34051) have the highest instability within the class ($D = 0.32$ – 0.45). This gradient, reproduced from Tower [14], provides a fine-grained rank ordering within the HDAC class: Spearman $\rho = 0.83$ between direction instability and selectivity rank across these 6 compounds after toxicity-gene correction. The HP1 “doesn’t bite” threshold ($\rho \geq 0.83$) was calibrated to this value.

Perturbation biology. CERES [8] corrects copy-number artifacts in CRISPR screens; integrated Broad-Sanger analyses [9] document cross-study divergence. These corrections operate on dependency-score magnitudes; our contribution is orthogonal, correcting subspace geometry and showing that rank-preserving corrections carry inferential value. The Connectivity Map framework [12, 5] and subsequent metric evaluations [6] establish the transcriptomic-similarity paradigm that direction instability extends. LEMUR [1] performs multivariate regression on Grassmann manifolds for multi-condition Perturb-seq; our Experiment 7 uses a simpler Grassmannian geodesic distance.

4 Methods

4.1 Data

Perturbation signatures were drawn from LINCS L1000 Phase I (GEO accession GSE92742; ~ 1.3 million profiles across ~ 50 cell lines), using Level 5 replicate-collapsed z-scores across 978 landmark genes [12]. We retained 8,949 small-molecule perturbagens tested in ≥ 5 cell lines.

Perturb-seq data derive from Replogle et al. [10]: genome-scale CRISPRi knockdowns in K562 and RPE1 cells. We extracted 1,676 genes with ≥ 30 perturbed cells in both cell types, computed per-gene expression responses over 2,073 highly variable genes, and fitted $k = 5$ PCA subspaces per gene per cell type (of these, 1,567 have matching DepMap Chronos labels for HP3; all 1,676 are used for HP1). Both screens target DepMap-essential genes; 85% of RPE1 targets overlap with K562 targets. Essentiality labels derive from DepMap Chronos scores [3]: 855 genes with Chronos < -0.5 in both K562 and RPE1 were classified as universal-essential; the remaining 712 are sub-threshold and serve as the comparison group. A label-permutation null (10,000 permutations) tested whether the observed group difference exceeds background

K562/RPE1 divergence.

JUMP-CP Cell Painting profiles derive from the Cell Painting Gallery [2]: 25,254 compounds tested in ≥ 5 imaging sources after filtering (78% attrition from the 116,750 total; the majority of compounds appear in fewer than 5 sources, making direction instability—which requires multiple contexts—undefined for them). Cell-health features (743 features matching Area, Compactness, Eccentricity, DNA/Mito Intensity, Granularity, Radial Distribution, and Texture Entropy) were identified by substring matching on CellProfiler feature names; corrected direction instability was computed on the remaining 2,437 features.

All scoring functions and drug lists were committed before any real LINCS signatures were analyzed (initial freeze: commit 8d74bd0). Perturb-seq hypotheses and extended drug hypotheses (H5-full, H3-CRISPRi) were pre-registered before the correction analysis (commit f3c88ed, tag prereg-extended-psb, timestamped 2026-07-08 11:56:58 UTC, preceding experiment execution at $\sim 12:00$ – $12:15$ UTC). The repository was public at the time of both freezes.¹

We treat the five LINCS hypotheses (H1–H5) as one test family and the three Perturb-seq hypotheses (HP1–HP3) as a second, because the two domains use different data sources, perturbation types, and geometric estimands (vector cosines vs. Grassmannian geodesics), making a joint family overly conservative. After Holm–Bonferroni correction within the drug family, H3 ($p < 10^{-27}$) and H5 ($p_{\text{Wilcoxon}} = 8.2 \times 10^{-13}$; also $p_{\text{perm}} < 10^{-3}$ on the preliminary 5-fold analysis) survive at $\alpha = 0.05$; H2 ($p = 0.28$) and H4 (gap = 0.018) do not; H1’s criterion-pass is an artifact. Within Perturb-seq, HP3 ($p_{\text{perm}} < 10^{-4}$) survives.

4.2 Phenotype-projected bracket (Experiment 3)

For each drug with an annotated target gene (Drug Repurposing Hub, 1,551 drugs), the “on-target direction” is the consensus shRNA knockdown signature of that target gene in LINCS L1000 (4,369 genes eligible, ≥ 3 hairpins each, consensus averaged across 20 cell lines). Phenotype-projected bracket projects the instability vector onto this on-target direction. On-target enrichment is quantified by cosine squared between each drug’s mean signature and its target gene’s shRNA consensus.

shRNA signatures have well-documented off-target effects, which add noise to the on-target direction. The shRNA consensus averages across ~ 20 cell lines, producing a cell-type-agnostic on-target direction at the cost of off-target contamination. A convergent-validity check using CRISPRi signatures (single cell type, cleaner loss-of-function but cell-type-specific) is reported in the Results.

The pre-registered prediction (H3) required Spearman $|\rho| > 0.3$ between phenotype-projected bracket and on-target enrichment, and $|\rho| < 0.15$ for raw bracket versus the same enrichment.

4.3 Transport-stable bracket (Experiment 5)

Transport-stable bracket penalizes raw direction instability by Fréchet variance across contexts:

$$\text{TS} = D - \lambda \cdot \text{Var}_{\text{Fréchet}} \quad (2)$$

where $\text{Var}_{\text{Fréchet}} = \frac{1}{K} \sum_i \arccos^2(\hat{\mathbf{s}}_i^\top \bar{\mathbf{s}})$ and $\bar{\mathbf{s}}$ is the Fréchet mean direction. The subtraction is a heuristic combination: the alternative (a ratio) would be undefined when $\text{Var}_{\text{Fréchet}} = 0$ and would amplify noise for highly consistent drugs. The penalty weight λ was fixed at the value from Tower [14] and was not optimized for this evaluation.

We evaluate via leave-one-cell-line-out cross-validation over all cell lines with ≥ 20 eligible drugs after excluding the held-out cell line (66 valid folds out of 71 total; 5 excluded for insufficient drug count). A preliminary 5-fold analysis using alphabetically-selected cell lines (A375, A549, A673, AGS, ASC) established the rare/common regime distinction with permutation nulls; the full leave-one-out extends this to all cell lines and adds Wilcoxon signed-rank as a co-primary criterion.

The pre-registered prediction (H5) required TS to outpredict raw bracket in $\geq 80\%$ of valid folds (co-primary 1), with Wilcoxon signed-rank on per-fold $\Delta\rho$ rejecting H_0 : median ≤ 0 at $\alpha = 0.05$ (co-primary 2).

¹Code, pre-registration, deviation log, and all experimental scripts: <https://github.com/elliotttower/direction-instability-drug-validity>. LINCS data: GEO GSE92742; Perturb-seq: Replogle et al. [10]; JUMP-CP: Chandrasekaran et al. [2]; DepMap: Dempster et al. [3].

4.4 Essential-gene correction on Perturb-seq (Experiment 7)

We computed the Fréchet mean subspace of the 855 essential-gene perturbation responses (separately in K562 and RPE1) via iterative projection on $\text{Gr}(5, 2073)$ (100 iterations, convergence to $\Delta < 10^{-6}$ in Grassmannian distance; the result is insensitive to initialization because the essential-gene subspaces cluster tightly). We then projected this essential-response subspace out of each gene’s response subspace, re-orthogonalized, and recomputed geodesic distances.

Two pre-registered predictions were tested. HP1 required Spearman ρ between raw and corrected distances below 0.70 (“bites”) or ≥ 0.83 (“doesn’t bite”). HP3 predicted that universal-essential knockdowns produce higher geodesic distances than non-essential knockdowns (Cohen’s $d > 0$).

4.5 Cell-health correction on JUMP-CP (Experiment 8)

The 743 cell-health features constitute 23% of the 3,180-feature space—proportionally six times larger than the LINCS toxicity gene set (41/978, 4%) and the largest feature removal in this study. If cell-health variance dominates cross-source consistency, this correction should reorganize the ranking more substantially than the drug or genetic corrections. This makes Experiment 8 the strongest test of the thesis: if corrections don’t bite even when the removed feature set is proportionally large, the ranking is robust.

The pre-registered prediction used the same threshold: Spearman ρ between raw and corrected direction instability below 0.70 (“bites”) or ≥ 0.83 (“doesn’t bite”).

4.6 Supplementary experiments

Three additional drug-domain experiments are reported in the Additional files (available in the code repository). **Experiment 1** (H1, toxicity correction) tests whether cytotoxic drugs masquerade as mechanism-conserving. **Experiment 2** (H2, broad-mechanism drugs) tests whether broad-target therapeutics fall below the population median. **Experiment 4** (H4, localization score) tests whether gene-set restriction improves MOA prediction. HP2 (GO pathway prediction from corrected Perturb-seq distances) is also in the Additional files.

5 Results

5.1 Pre-registered scorecard

Table 1 summarizes all pre-registered hypotheses. Two of five drug hypotheses survive Holm–Bonferroni correction (H3, H5); H2 meets its pre-registered criterion but does not survive multiplicity correction; H1 is refuted with an artifact criterion-pass; H4 is an informative null. In the Perturb-seq domain, HP3 is confirmed; HP1 shows the correction “doesn’t bite” at the population level; HP2 is a null. Experiment 8 (JUMP-CP) shows the correction “doesn’t bite.”

5.2 Experiment 3: Phenotype-projected bracket separates on-target from off-target instability

Among 795 drugs with annotated targets and matched shRNA consensus signatures, phenotype-projected bracket correlates with on-target enrichment at Spearman $\rho = 0.376$ ($p = 4.8 \times 10^{-28}$). Raw bracket shows no correlation ($\rho = -0.043$, $p = 0.22$). Both pre-registered criteria are satisfied.

H3 confirms Rung 5 of the validity ladder. Raw direction instability measures whether a drug’s signatures point in consistent directions, but this consistency could arise from any coherent program. Projecting onto the genetic perturbation direction of the annotated target isolates the mechanistically relevant component. The near-zero raw correlation shows that global instability carries no on-target information; the strong projected correlation shows that the on-target projection recovers it.

The gene-set construction avoids researcher degrees of freedom: the shRNA consensus is an empirical object from the same LINCS platform, requiring no pathway database selection. H3 and H4 share this shRNA dependency (documented in DEVIATION.LOG.md entry 3) and are not independent tests. The

Table 1: Pre-registered scorecard. Hypotheses are grouped by domain. “Holm” indicates survival after Holm–Bonferroni correction within the drug family ($\alpha = 0.05$). Experiment references are to the full methods in the body (Exp. 3, 5, 7, 8) or Additional files (Exp. 1, 2, 4). Result details for starred entries appear in Section 5.6.

	Prediction	Result	Outcome	Holm
H1*	Toxicity reclassifies	$\Delta < 0.01$	Refuted	—
H2*	Broad drugs below median	7/10 below	Confirmed	No
H3	$ \rho_{\text{proj}} > 0.3$	$\rho = 0.376$	Confirmed	Yes
H4*	AUROC gap > 0.05	Gap = 0.018	Null	No
H5	TS wins $\geq 80\%$ folds	66/66 folds, $p_W = 8.2 \times 10^{-13}$	Confirmed	Yes
HP1	Correction bites ($\rho < 0.70$)	$\rho = 0.91$	Doesn’t bite	—
HP2*	GO prediction improves	$\Delta\rho = -0.02$	Doesn’t bite	—
HP3	Essential $d > 0$	$d = 0.60$	Confirmed	Yes
Exp. 8	Health correction bites	$\rho = 0.987$	Doesn’t bite	—

phenotype-projected bracket is applicable only to drugs with target annotations—1,551 of 8,949 (17%). For the remaining 82% of drugs with unknown targets, the validity ladder provides other rungs (transport stability, localization) that do not require target annotations.

As a pre-registered convergent-validity check, we recomputed phenotype-projected bracket using CRISPRi loss-of-function signatures [10] as the on-target direction. The projected-bracket correlation with on-target enrichment did not reach significance (Spearman $\rho = -0.12$, $p = 0.16$, $n = 131$), corresponding to pre-registered outcome (c). On the 114-drug matched subset (drugs with both ground truths), shRNA projection yields $\rho = 0.42$ while CRISPRi projection yields $\rho = -0.14$, confirming that the ground-truth source—not the drug set—drives the difference.

The pre-registered secondary criterion points to the likely cause: raw bracket correlated positively with CRISPRi enrichment ($\rho = +0.27$, $p = 0.002$) while projected bracket did not—the reverse of the shRNA pattern. This reversal is consistent with the pre-registered K562-only confound: CRISPRi signatures derive from a single hematopoietic cell type, so the CRISPRi “on-target direction” captures cell-type-specific effects rather than cross-context mechanism—a confound that multi-cell-line shRNA consensus avoids by averaging across ~ 20 cell lines. One plausible explanation for the positive raw-bracket/CRISPRi correlation, which we did not test directly, is that K562 expresses many drug targets at high levels, producing spurious alignment between raw instability and K562-specific signatures.

CRISPRi does not provide independent convergent validity for H3. The shRNA-based result stands on its original evidence rather than being reinforced.

5.3 Experiment 5: Transport-stable bracket outpredicts raw bracket in held-out cell lines

Transport-stable bracket outpredicts raw bracket in all 66 valid leave-one-out folds. Both co-primary criteria are satisfied: TS wins 66/66 folds (100%, criterion $\geq 80\%$) and Wilcoxon signed-rank on per-fold $\Delta\rho$ yields $p = 8.2 \times 10^{-13}$ (criterion $p < 0.05$).

Across 66 folds, mean raw $\rho = -0.60$ (anti-predictive in every fold; range $[-0.78, -0.22]$) while mean TS $\rho = +0.50$ (positively predictive). Mean $\Delta\rho = +1.10$ (95% bootstrap CI $[1.00, 1.19]$; all 66 values positive, range $[0.17, 1.57]$). Five cell lines were excluded for having < 20 eligible drugs. Figure 4 shows the $\Delta\rho$ distribution and the frequency gradient; Table 2 shows representative folds illustrating the rare/common regime distinction.

In rarer folds (such as A673, AGS, ASC), raw bracket is strongly anti-predictive ($\rho = -0.37$ to -0.73): drugs with high raw instability in training produce held-out signatures that point *away* from the training consensus. Transport-stable bracket reverses this, becoming positively predictive ($\rho = +0.35$ to $+0.73$).

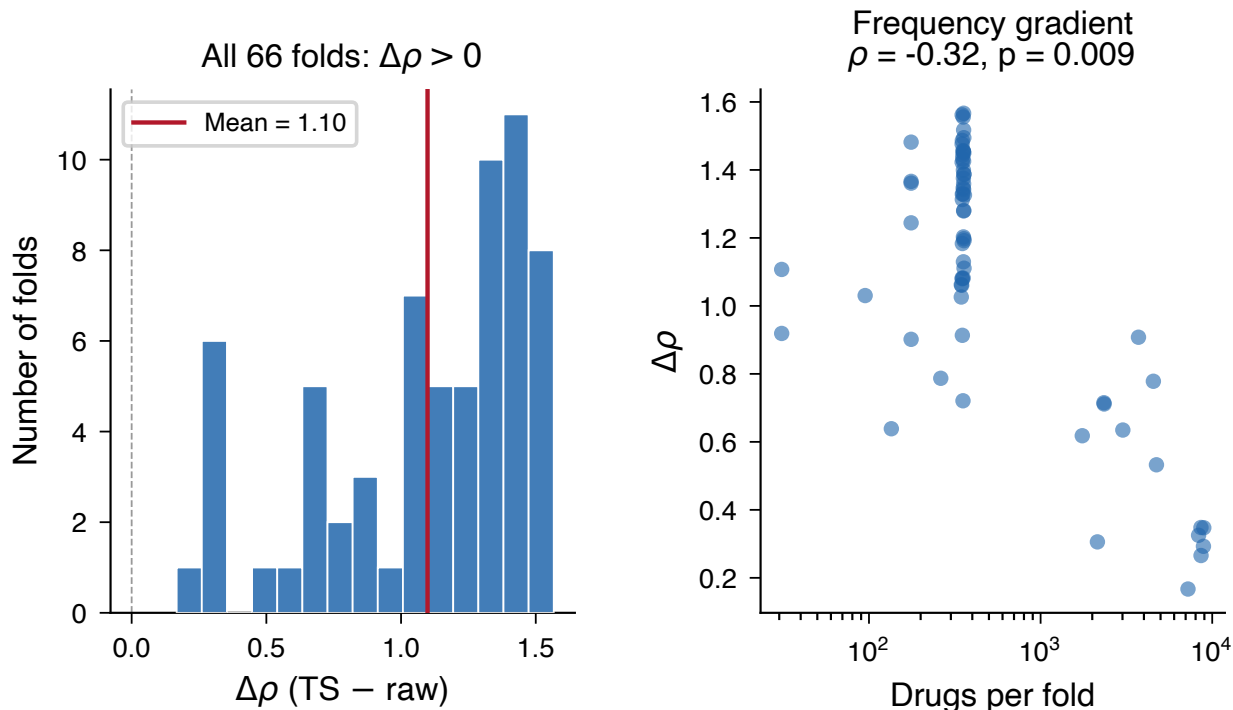


Figure 4: **Full 66-fold leave-one-out: TS advantage is universal.** **Left:** Distribution of per-fold $\Delta\rho$ (TS - raw Spearman correlation with held-out cosine). All 66 values are positive; the mean (red line) is +1.10. **Right:** Rarer cell lines (fewer drugs per fold) show larger TS advantages (Spearman $\rho = -0.32, p = 0.009$), confirming that the rare/common regime distinction from the preliminary 5-fold analysis generalizes across all cell lines.

The reversal occurs because removing a single rare cell line substantially changes the consensus direction; drugs whose training instability was driven by that cell line are precisely those that TS downweights via the Fréchet penalty. This pattern holds across all 66 folds: the frequency-improvement correlation (Spearman $\rho = -0.32, p = 0.009$) confirms that rarer cell lines show larger TS advantages.

In common folds (8,000+ drugs), both metrics remain anti-predictive. TS improves over raw ($\rho = -0.155$ vs. -0.48 for A375; -0.201 vs. -0.49 for A549), but both correlations stay negative. TS “wins” in these folds by losing less badly, not by becoming positively predictive. The consensus direction is stable when computed from many cell lines, leaving less room for the Fréchet penalty to improve prediction.

Raw direction instability is anti-predictive because it conflates two sources of angular consistency: genuine mechanism conservation (signal) and the geometric tightness of the consensus direction around a small number of contexts (noise). Drugs with low raw DI in training tend to have tight consensus directions that are sensitive to removing any single cell line—precisely the drugs that fail to predict held-out effects. This is a failure of the raw estimator when applied to leave-one-out prediction without variance correction, not a failure of direction instability as a concept.

Before conducting the full 66-fold leave-one-out, we tested whether the TS advantage could be algebraic using a variance-preserving permutation null on the preliminary 5-fold analysis. Held-out *signatures* are shuffled across drugs while training data (raw DI, Fréchet variance, TS, consensus vector) remain fixed, and held-out cosines are recomputed against each drug’s original consensus. Under this null (1,000 permutations), TS wins only 0.1 of 5 folds on average with mean $\Delta\rho = -0.12$ (raw DI *outpredicts* TS when drug-specific signal is broken). The observed advantage falls entirely outside the null distribution ($p_{\text{perm}} < 10^{-3}$). The full 66-fold Wilcoxon test ($p = 8.2 \times 10^{-13}$) provides independent confirmation that the advantage is drug-specific.

Table 2: Leave-one-cell-line-out cross-validation (selected folds from the full 66-fold analysis). Transport-stable (TS) bracket outpredicts raw bracket in all folds. In common folds (8,000+ drugs), both metrics are anti-predictive but TS is less so. In rarer folds, TS reverses the sign entirely. Full 66-fold results: mean $\Delta\rho = +1.10$, Wilcoxon $p = 8.2 \times 10^{-13}$.

Held-out cell	N_{drugs}	ρ_{raw}	ρ_{TS}	Winner
A375	8,328	-0.480	-0.155	TS
A549	8,888	-0.494	-0.201	TS
A673	357	-0.725	+0.725	TS
AGS	356	-0.712	+0.714	TS
ASC	2,345	-0.365	+0.346	TS

5.4 Experiment 7: Essential-gene correction preserves ranking but reveals confound-dominated machinery

Universal-essential knockdowns have *higher* Grassmannian geodesic distances than non-essential knockdowns (mean 2.80 vs. 2.67, Cohen’s $d = +0.60$, Mann–Whitney $p < 10^{-33}$; $n = 1,567$ genes with both geodesic distances and DepMap labels, of which 855 universal-essential and 712 sub-threshold). Cohen’s $d = 0.60$ is a medium effect—the distributions overlap substantially, and individual genes span the full range in both groups. The finding is that the *group means* differ in the opposite direction from the predicted confound, not that essentiality status determines geodesic distance.

Essential genes do not produce a shared response across cell types; they *diverge*—the same gene’s lethal disruption manifests through different transcriptomic programs in K562 and RPE1 (Figure 5). This is consistent with cell-line-specific essentiality effects documented in CERES [8] and integrated Broad–Sanger analyses [9]: the functional consequences of losing a core-machinery gene depend on the cell’s lineage and metabolic context.

A label-permutation null (shuffling essential/non-essential labels across the 1,567 genes, 10,000 permutations) confirms that this divergence exceeds background K562/RPE1 divergence: the observed difference (0.123) falls outside the null 95% interval $[-0.021, 0.022]$, $p_{\text{perm}} < 10^{-4}$. The null preserves the geodesic-distance distribution while breaking only the label assignment, so the $\sim 6\times$ exceedance of the null half-width rules out the library-overlap confound (85% target overlap between screens).

Spearman ρ between raw and corrected geodesic distances is 0.91 ($p < 10^{-100}$). The global ranking is preserved.

The correction’s value is at the item level: individual genes shift hundreds of rank positions. The top 100 rank-shifters are enriched for spliceosomal complex assembly (Fisher’s exact OR = 8.9, $p = 1.3 \times 10^{-7}$) and proteasomal catabolic processes (OR = 4.1, $p = 6.7 \times 10^{-5}$). Proteasome subunits (*PSMA3*, *PSMB6*, *PSMC4*) shift 750–900 positions; spliceosome components (*SF3B1*, *SNRPG*, *DDX23*) shift 660–820 positions. These core-machinery complexes are the genes whose cross-cell-type geometry is most dominated by the shared essential-response axis. The correction provides a concrete action: flag these genes before interpreting their transport scores.

5.5 Experiment 8: Cell-health correction has negligible effect in JUMP-CP

Among 25,254 compounds, corrected direction instability tracks raw at Spearman $\rho = 0.987$ ($p < 10^{-100}$). By the pre-registered criterion, this “doesn’t bite”—the strongest result for ranking robustness across all three domains.

The 743 health features constitute 23% of the 3,180-feature space—proportionally six times larger than the drug toxicity gene set (41/978, 4%)—yet produce *less* ranking impact ($\rho = 0.987$ vs. $\rho > 0.99$ for LINCS). That a proportionally larger feature removal barely moves the ranking is stronger evidence for the thesis than the drug case: cell-health features carry less of the cross-source variance in morphological signatures than stress genes carry in transcriptomic signatures. The 78% attrition from the full JUMP-CP dataset (116,750

to 25,254 compounds) means the retained set is biased toward well-replicated compounds; the result may not generalize to sparsely-profiled compounds.

One explanation for the negligible effect: in morphological space, “health” and “mechanism” features are less separable than in transcriptomic space. A drug that alters cell shape changes both health-associated features (area, compactness) and mechanism-specific features (texture, radial distribution) simultaneously, because morphological effects are inherently coupled. In 978-gene transcriptomic space, stress-response genes and pathway-specific genes are more separable, which is why the LINCS toxicity correction has a larger (though still small) effect.

Despite near-perfect global correlation, individual compounds shift up to 7,155 rank positions, identifying those whose apparent morphological consistency is health-feature-dominated. The cell-health feature definition (substring matching on CellProfiler names) is a limitation: a partition based on correlation with viability assays would be more principled and could yield different correction magnitudes.

5.6 Summary of supplementary experiments

Full methods, tables, and discussion appear in the Additional files.

H1 (toxicity correction). Among 13 cytotoxic drugs, most are already high-instability ($D > 0.87$ for 9 of 13); toxicity correction shifts values by < 0.01 . The predicted failure mode—cytotoxicity masquerading as mechanism conservation—does not hold. Cytotoxicity produces genuinely context-dependent signatures in 978-gene space.

H2 (broad-mechanism drugs). Seven of 10 broad-mechanism drugs fall below the population median. Pan-HDAC inhibitors dominate ($D = 0.49$ – 0.55). The pre-registered second prediction—that kinase inhibitors would fall above the median—failed. The sorting axis is target breadth, not clinical specificity.

H4 (localization). Macro-averaged AUROC gap is $+0.018$ (below $+0.05$ threshold). Localization improves prediction for intracellular enzyme inhibitors (topoisomerase: $+0.49$, CDK: $+0.26$) but hurts for receptor-mediated drugs (acetylcholine: -0.20 , PPAR: -0.19). Rung 4 is domain-dependent.

HP2 (pathway function prediction). GO Jaccard overlap has near-zero correlation with geodesic distance ($|\rho| < 0.025$) for both raw and corrected forms. Geometric similarity of perturbation responses is distinct from functional pathway relatedness.

6 Discussion

6.1 Construct validity separates score from claim

The central finding is negative in the most useful sense: validity conditions do not reorganize the direction-instability ranking. This thesis was genuinely at risk: had any correction produced $\rho < 0.70$ while simultaneously improving prediction of an external outcome, the “reclassify” interpretation would have prevailed—the raw ranking would have been wrong, not merely under-warranted. That no correction crossed this threshold across three domains is the empirical result. The Perturb-seq result involves a degenerate $K = 2$ pairwise comparison rather than genuine $K \geq 5$ angular consistency; the qualitative conclusion holds across domains, but the genetic domain tests a weaker geometric estimand.

A rank-preserving correction (“doesn’t bite”) admits two interpretations: the targeted confound is absent, or the confound is present but rank-preserving by construction. These are distinguishable when item-level rank shifts are examined: if the correction produces large individual shifts (such as proteasome/spliceosome genes shifting 660–900 positions in Perturb-seq) while preserving the global correlation, the confound is real but concentrated in identifiable items. The population-level ρ and the item-level rank shifts together provide richer evidence than either alone.

What changes is the inferential warrant. Phenotype projection (H3) shows that direction instability decomposes into on-target and off-target components ($\rho = 0.38$ vs. -0.04); a CRISPRi convergent-validity check does not independently confirm this finding, likely because K562-only CRISPRi signatures are confounded by cell-type-specific effects. Transport stability (H5) shows the metric predicts held-out effects only after Fréchet variance correction; raw bracket is anti-predictive in all 66 folds. Essential-gene correction identifies proteasome/spliceosome genes as confound-dominated while the population ranking holds.

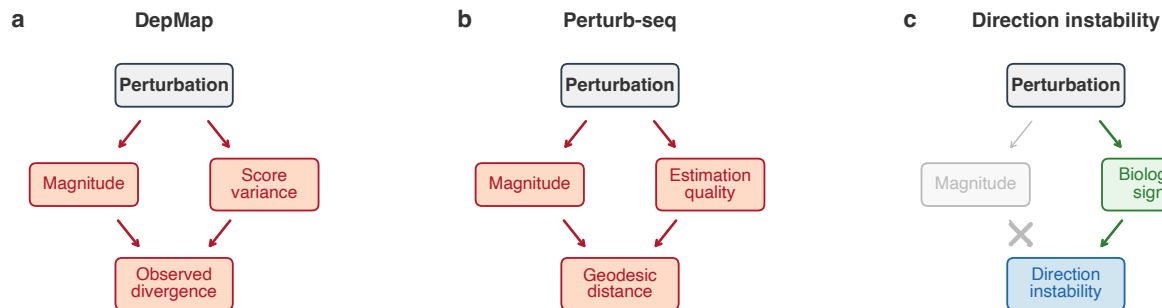


Figure 5: **Cross-domain confound structures.** In all three domains, magnitude confounds operate through different mechanisms, but cosine normalization renders direction instability (right panel) invariant to perturbation magnitude. **Left:** DepMap variance-based scores conflate essentiality label with score variance. **Center:** Perturb-seq estimation-based metrics conflate essentiality status with estimation quality. **Right:** Cosine normalization renders direction instability invariant to perturbation magnitude, so the observed association reflects biological signal. Despite different confound mechanisms across domains, the validity corrections produce the same qualitative result: population rankings are preserved ($\rho > 0.83$) while item-level shifts identify specific confound-dominated perturbations.

“Warrant upgrade” is a concrete claim about downstream decisions, not a philosophical abstraction. Consider two scenarios:

Drug repurposing triage. Two drugs both have $D = 0.55$. Drug X has passed Rungs 4–6: its low instability is localized to the target pathway (H3-type projection confirms on-target alignment), and it replicates across held-out cell lines (H5-type cross-validation passes in 66/66 folds). Drug Y has only raw $D = 0.55$ —no additional validation. A practitioner prioritizes Drug X for experimental follow-up because Drug Y’s raw score could reflect off-target consistency (H3 shows raw bracket carries no on-target information) or sampling noise (H5 shows raw bracket is anti-predictive). The warrant upgrade changes the follow-up priority.

Perturb-seq hit triage. Among 100 genes with low geodesic distance, 15 shift >500 rank positions after essential-gene correction (proteasome, spliceosome components). These are flagged as potentially confound-dominated. The remaining 85 genes have the same transport scores but stronger warrant—they are prioritized for functional validation. The warrant upgrade changes which genes proceed to follow-up.

6.2 Cross-domain consistency

The same geometric object—angular deviation of perturbation-response directions across contexts—captures mechanism conservation in three unrelated domains (Table 3). All three exhibit the same pattern: the relevant validity condition preserves the global ranking while identifying individual perturbations whose scores are most confound-dominated.

The domain-specific failure modes differ: in drugs, cytotoxicity produces high-instability signatures; in genetics, universal-essential knockdowns produce high-distance subspaces (divergent lethal programs); in morphology, cell-health features contribute a small fraction of cross-source variance. All three invert the naive prediction that “violence mimics transport.”

6.3 Generalizability of the construct-validity procedure

The validity ladder was tested on one geometric perturbation score (direction instability) across three biological domains. The *procedure*—define interpretive threats, specify corrections that target each threat, pre-register predictions about whether corrections reorganize rankings, test empirically—applies to other geometric scores that measure perturbation-response consistency, such as connectivity scores [12], morphological similarity indices, and Grassmannian distance metrics. What would change are the domain-specific rungs: a connectivity score would require different confound corrections than direction instability, but the

Table 3: Cross-domain summary. All three domains exceed the “doesn’t bite” threshold ($\rho \geq 0.83$). The LINCS HDAC gradient ρ measures fine-grained ordering within a single drug class (6 compounds in the selectivity gradient, Figure 3); the global 8,949-drug $\rho > 0.99$. H5 (transport stability) tests predictive validity rather than rank preservation; the 66-fold leave-one-out result (Table 1) provides the strongest domain-specific evidence for LINCS.

Domain	Correction	N	ρ
LINCS (HDAC gradient)	Toxicity genes	6	0.83
LINCS (global)	Toxicity genes	8,949	>0.99
Perturb-seq	Essential-gene subspace	1,676	0.91
JUMP-CP	Cell-health features	25,254	0.99

same structure of threat identification, correction, and pre-registered testing applies. The effect size for phenotype projection ($\rho = 0.38$, explaining $\sim 14\%$ of variance) is moderate; contextualizing this against other drug-mechanism prediction methods would help practitioners judge whether this rung’s evidence is sufficient for their decisions.

6.4 Scope and limitations

Direction instability measures angular consistency of perturbation signatures across contexts. It does not measure therapeutic efficacy, in-vivo translatability, or clinical specificity. The validity ladder is a construct-validation framework for interpretive warrant, not a surrogate endpoint.

The drug experiments H3–H5 depend on shRNA knockdown signatures as ground-truth on-target directions. shRNA has well-documented off-target effects, which add noise to the ground truth. A CRISPRi convergent-validity check did not independently confirm the phenotype-projection result ($\rho = -0.12$, $p = 0.16$, $n = 131$); the matched 114-drug comparison (shRNA $\rho = 0.42$ vs. CRISPRi $\rho = -0.14$) points to the pre-registered K562-only confound (CRISPRi signatures from a single cell type are poor on-target directions for drugs acting in non-hematopoietic contexts). The shRNA H3 result is not refuted but neither is it reinforced. Drugs with unannotated targets (7,398 of 8,949) are excluded from H3 and H4, limiting these rungs to 17% of the dataset. LINCS measures 978 landmark genes, restricting the toxicity gene set to the landmark panel.

The Perturb-seq analysis uses $K = 2$ cell types, providing no Fréchet variance, holonomy, or transport stability test. Both screens target DepMap-essential genes, so the HP3 “non-essential” group consists of sub-threshold essentials rather than genuinely non-essential genes. A genome-wide screen with genuine non-essential controls would be sharper, though the label-permutation null rules out the library-overlap confound.

The transport-stable bracket formula (Eq. 2) is a heuristic linear combination of direction instability and Fréchet variance, not a principled geometric operation on the signature manifold. The penalty weight λ was inherited from the companion paper [14] and was not optimized for this evaluation. A sensitivity analysis across a range of λ values would strengthen the claim that the H5 advantage is robust to the specific penalty weight; we report this specific formula because it was pre-registered, and post-hoc λ optimization would compromise the pre-registration. A ratio or a geodesic-based combination might perform differently.

The validity ladder draws an analogy to Bradford Hill criteria for causal inference. Bradford Hill criteria are individually neither necessary nor sufficient for establishing causation; they are heuristic guidelines. The validity ladder shares this character: it is a structured set of proof obligations rather than a formal deductive system. Each rung eliminates a specific threat, but satisfying all tested rungs does not *guarantee* that a mechanistic claim is correct—it guarantees that specific alternatives have been ruled out.

JUMP-CP cell-health features were defined by substring matching rather than by correlation with an independent viability assay. The 78% attrition from the full dataset biases the retained set toward well-replicated compounds.

7 Conclusion

We have proposed and tested a construct-validity framework for geometric perturbation scores, using direction instability as a case study across three biological domains (drug transcriptomics, genetic perturbation subspaces, morphological profiles). Validity conditions do not improve the score’s ability to rank perturbations—the ranking is already biologically meaningful in all three domains ($\rho > 0.99$ for LINCS, 0.91 for Perturb-seq, 0.99 for JUMP-CP after correction). They improve the practitioner’s ability to make claims about that ranking and to make decisions based on those claims.

The procedure tested here—threat identification, correction specification, pre-registered prediction, empirical test—is applicable to any geometric score measuring perturbation-response consistency across contexts. Phenotype projection separates on-target from off-target consistency ($\rho = 0.38$ vs. -0.04); variance correction outpredicts the raw score in all 66 leave-one-out folds ($p_{\text{Wilcoxon}} = 8.2 \times 10^{-13}$, mean $\Delta\rho = +1.10$); localization helps for intracellular enzyme targets but not receptor-mediated drugs, constraining where specific rungs apply. The domain-specific failure modes differ, but the pattern is the same: correction preserves population rankings while identifying individual perturbations most susceptible to confounding. The validity ladder provides a structured procedure for accumulating evidence that a geometric perturbation score licenses mechanistic inference.

Declarations

Ethics approval and consent to participate

Not applicable. This study uses only publicly available datasets (LINCS L1000, Perturb-seq, JUMP-CP, DepMap) with no human subjects research.

Consent for publication

Not applicable.

Availability of data and materials

All data used in this study are publicly available: LINCS L1000 (GEO GSE92742), Perturb-seq [10], JUMP-CP [2], DepMap [3]. Code, pre-registration commits (SHA 8d74bd0 for initial freeze; SHA f3c88ed, tag `prereg-extended-psb`, for extended pre-registration), deviation log, and all experimental scripts are available at <https://github.com/elliotttower/direction-instability-drug-validity>. The repository was public before the pre-registration freeze.

Competing interests

The author declares no competing interests.

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Authors’ contributions

ET conceived the study, designed the validity framework, conducted all experiments, and wrote the manuscript.

Use of AI-assisted technologies

Claude (Anthropic) was used as a writing and analysis assistant during manuscript preparation. The author directed all analytical decisions, verified all numerical results against source data, and takes full responsibility for the content. The validity framework, pre-registered hypotheses, all experimental results, and interpretations are the author’s own. No AI tool is listed as an author.

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Companion papers

Direction instability and the transport-stable bracket formula are defined and validated in a companion paper on LINC S L1000 drug transport (submitted to PLOS Computational Biology, PCOMPBIOL-S-26-02155; [14]). The cross-domain direction-instability atlas (LINC S, Perturb-seq, JUMP-CP) is described in a companion paper submitted to GigaScience (GIGA-D-26-00229; [13]). Both are available as Zenodo preprints. The present paper tests construct-validity conditions on top of these prior results.

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